Final Exam

● Graded

Student

Hengkai Yu

Total Points

45 / 45 pts

Question 1

Overall Grades **45** / 45 pts

✔ **– 0 pts** See the details on each problem.

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Hengkai Yu	hy266	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- **No Cellphones.**
- **No Calculators.**
- **Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.**
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE:

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	5	5	5	45

Problem 1. (5) Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

coefficients matrix : $A = \begin{pmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 1 & 0 & 2 \end{pmatrix}$

Augmented matrix : $\begin{pmatrix} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{pmatrix}$ $R_3 \to R_3 - \frac{1}{2}R_1$ $\sim$ $\begin{pmatrix} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & \frac{3}{2} & 2 & \frac{1}{2} \end{pmatrix}$

$R_1 \to R_1 - \frac{3}{2}R_2$, $R_3 \to R_3 + \frac{3}{4}R_2$ $\sim$ $\begin{pmatrix} 2 & 0 & \frac{9}{2} & -\frac{1}{2} \\ 0 & -2 & -3 & 1 \\ 0 & 0 & -\frac{1}{4} & \frac{5}{4} \end{pmatrix}$

$R_1 \to R_1/2$, $R_2 \to R_2/-2$, $R_3 \to R_3 \cdot (-4)$ $\sim$ $\begin{pmatrix} 1 & 0 & \frac{9}{4} & -\frac{1}{4} \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -5 \end{pmatrix}$

Therefore, $\operatorname{rank}(A) = 3 = n$, no free variable, one solution.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$$

Therefore, general solution is $\begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$.

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) Row reduction for A :

$\begin{pmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{pmatrix}$ $R_1 \to R_1/2$, $R_2 \to R_2/-5$ $\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{pmatrix}$ $R_2 \to R_2 - R_1$, $R_4 \to R_4 - R_1$ $\begin{pmatrix} 1 & 2 & 1 \\ 0 & -2 & -2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$

$R_2 \to R_2/-2$, $R_1 \to R_1 - R_2$, $R_4 \to R_4 - R_2$ $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ two pivots → two pivot columns, in C_1, C_2

Therefore, a basis of the column space of A is $\left\{ \begin{pmatrix} 2 \\ -5 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 0 \\ 1 \\ 3 \end{pmatrix} \right\}$

(2) Two pivot columns are on Row 1, Row 2,

Therefore, a basis of the row space of A is $\{(2, 4, 2), (-5, 0, 5)\}$

(3) $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} X = 0$. $X = \begin{pmatrix} 5 \\ -5 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} 5$, nullity $= n - r = 3 - 2 = 1$.

Therefore, a basis for nullspace is $\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$.

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$$A\begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix} = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}$$

Let X be $\begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix}$. X is clearly full rank, so invertible.

$\det(X) = -1 \neq 0$

$AX = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}$, so $AXX^{-1} = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}X^{-1}$.

Therefore $A = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}X^{-1}$

To find X^{-1}:

$$\left(\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right) \xrightarrow{R_1 \to R_1 + R_2,\ R_3 \to R_3 + R_2} \left(\begin{array}{ccc|ccc}1&1&0&1&1&0\\0&-1&-1&0&1&1\\-1&-1&-1&0&0&1\end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 + R_1} \left(\begin{array}{ccc|ccc}1&1&0&1&1&0\\0&-1&-1&0&1&1\\0&0&-1&1&1&1\end{array}\right) \xrightarrow{R_1 \to R_1 + R_2,\ R_2 \to -R_2} \left(\begin{array}{ccc|ccc}1&0&-1&1&2&1\\0&1&1&0&-1&-1\\0&0&-1&1&1&1\end{array}\right)$$

$$\xrightarrow{\substack{R_1 \to R_1 - R_3\\ R_2 \to R_2 - R_3}} \left(\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right)$$

Therefore, $X^{-1} = \begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix}$

So, $A = \begin{pmatrix}1&0&2\\1&2&2\end{pmatrix}\begin{pmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{pmatrix} = \begin{pmatrix}-2&-1&-2\\0&-1&-2\end{pmatrix}$

Problem 4. [5] Find the Jordan canonical form of

Let $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix} = A$

$A - \lambda E = \begin{pmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{pmatrix}$ $p(\lambda) = (1-\lambda)^2(2-\lambda)$

$\lambda_1 = 1$. $AM = 2$. $(A-E)V_1 = 0$.

$$\begin{pmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{pmatrix} V_1 = 0. \quad \text{So } V_1 = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} \quad GM = 1 < 2.$$

$(A-E)V_2 = V_1$. $\begin{pmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{pmatrix} V_2 = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$. $V_2 = \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix}$

$\lambda_2 = 2$. $AM = 1$. $(A-2E)V_3 = 0$.

$$\begin{pmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{pmatrix} V_3 = 0. \text{ So } V_3 = \begin{pmatrix} 2l \\ l \\ -4l \end{pmatrix}$$

let $s = 1$, $t = 1$, $l = 1$. we have

$A = PJP^{-1}$, in which $J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. $P = \begin{pmatrix} 0 & -\frac{1}{2} & 2 \\ 0 & 0 & 1 \\ 1 & 1 & -4 \end{pmatrix}$.

$P^{-1} =$

$A = PJP^{-1} = \begin{pmatrix} 0 & -\frac{1}{2} & 2 \\ 0 & 0 & 1 \\ 1 & 1 & -4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 & 1 \\ -2 & 4 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

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Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$f(x,y,z) = X^T \begin{pmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{pmatrix} X$, let $\begin{pmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{pmatrix} = A$

$A - \lambda E = \begin{pmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{pmatrix}$. $\det(A-\lambda E) = (3-\lambda)^2(4-\lambda) - 2(3-\lambda)$

$= (3-\lambda)(\lambda^2 - 7\lambda + 12 - 2)$

$= (3-\lambda)(2-\lambda)(5-\lambda)$.

$\lambda_1 = 2$. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{pmatrix} X = 0$. $V_1 = \begin{pmatrix} -s \\ s \\ s \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} s = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$

$\lambda_2 = 3$. $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{pmatrix} X = 0$. $V_2 = \begin{pmatrix} t \\ t \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} t = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$

$\lambda_3 = 5$. $\begin{pmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{pmatrix} X = 0$. $V_3 = \begin{pmatrix} -l \\ l \\ -2l \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} l = \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{pmatrix}$

let ~~s, t, l~~ $s = \frac{1}{\sqrt{3}}$, $t = \frac{1}{\sqrt{2}}$, $l = \frac{1}{\sqrt{6}}$

$X^T A X = X^T P D P^{-1} X$, $P = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}$, $D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$

$= y^T D y$

$y = \begin{pmatrix} u \\ v \\ w \end{pmatrix} = P^{-1} X = P^T X = \begin{pmatrix} -\frac{1}{\sqrt{3}}x + \frac{1}{\sqrt{3}}y + \frac{1}{\sqrt{3}}z \\ \frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y \\ -\frac{1}{\sqrt{6}}x + \frac{1}{\sqrt{6}}y - \frac{2}{\sqrt{6}}z \end{pmatrix}$

$X^T A X = 2u^2 + 3v^2 + 5w^2$.

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

Solve :

$$\begin{pmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -3 \\ 8 \\ -8 \end{pmatrix}$$

Augmented matrix : $\left(\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right)$ $R_3 \to R_3/2$, $R_3 \leftrightarrow R_1$ $\left(\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \end{array}\right)$

$R_2 \to R_2 - 2R_1$ $\left(\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right)$ $R_3 \to R_3 - 2R_2$, $R_1 \to R_1 + R_2$ $\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right)$

rank (A) = 3. one solution.

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = z \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$$

Therefore $w = 3v_1 - v_2 + v_3$

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

Let $u_1 = v_1 = \begin{pmatrix}0\\1\\1\\0\end{pmatrix}$

$u_2 = v_2 - \frac{v_2^T u_1}{\|u_1\|^2} u_1$

$= \begin{pmatrix}3\\0\\2\\-1\end{pmatrix} - \begin{pmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{pmatrix} \begin{pmatrix}0\\1\\1\\0\end{pmatrix}$

$= \begin{pmatrix}3\\\frac{1}{2}\\\frac{3}{2}\\-1\end{pmatrix} \begin{pmatrix}3\\-1\\1\\-1\end{pmatrix}$

now we have orthogonal basis $\{u_1, u_2\}$, $u_1 \cdot u_2 = 0$

now normalize them:

$w_1 = \frac{u_1}{\|u_1\|} = \begin{pmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{pmatrix}$, $w_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix}\frac{3}{2\sqrt{3}}\\-\frac{1}{2\sqrt{3}}\\\frac{1}{2\sqrt{3}}\\-\frac{1}{\sqrt{3}}\end{bmatrix}$

Therefore, an orthonormal basis for V is: $\left\{\begin{bmatrix}0\\\frac{\sqrt{2}}{2}\\\frac{\sqrt{2}}{2}\\0\end{bmatrix}, \begin{bmatrix}\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{6}\\\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{2}\end{bmatrix}\right\}$

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Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$\det(A) = \det(A^T) =$ ~~de~~

Since $A^T = -A$

$\det(A) = \det(A^T) = \det(-A)$

So $\det(A) = \det(-A)$

Operate on A : time each row by (-1) and get $(-A)$

So $\det(-A) = (-1)^n \det(A)$

when n is odd, $\det(-A) = -\det(A)$

So $\det(A) = \det(-A) =$ ~~0~~ $-\det(A)$

Therefore $\det(A) = 0$

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-\vec{v}\|^2$ $\quad \left(\sqrt{(\vec{u}\pm\vec{v})(\vec{u}\pm\vec{v})} = \|\vec{u}\pm\vec{v}\| \right)$

$(\vec{u}+\vec{v})(\vec{u}+\vec{v}) = (\vec{u}-\vec{v})(\vec{u}-\vec{v})$

$\|\vec{u}\|^2 + \vec{u}\cdot\vec{v} + \vec{v}\cdot\vec{u} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - \vec{u}\cdot\vec{v} - \vec{v}\cdot\vec{u} + \|\vec{v}\|^2$

So $\quad \vec{u}\cdot\vec{v} = -\vec{v}\cdot\vec{u}$

∵ According to definition of dot product of vectors :

$\vec{u}\cdot\vec{v} = \vec{v}\cdot\vec{u}$

Therefore, $2\vec{u}\cdot\vec{v} = 0 \quad \vec{u}\cdot\vec{v} = 0$,

So $\vec{u}$ and $\vec{v}$ are perpendicular.